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Image capturing device and image calculation method

Abstract

An image capturing device with a sensor, a memory, and a processor is provided. The sensor captures an image of at least one target. The memory stores a plurality of instructions. The processor obtains the plurality of instructions to perform the following steps: controlling the sensor to capture a reference image and a processed image; capturing a first bright region and a dark region from the reference image, and capturing a second bright region from the processed image; performing calculations on a first brightness value of the first bright region and a second brightness value of the second bright region respectively with at least two first brightness thresholds, to obtain a first low exposure compensation value and a second low exposure compensation value; and obtaining a high exposure compensation value according to comparisons between a third brightness value of the dark region and at least two second brightness thresholds.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan Application Serial No. 110134961, filed on Sep. 17, 2021. The entirety of the above-mentioned patent application is hereby incorporated by

reference herein and made a part of the specification.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The disclosure relates to an image capturing device and an image calculation method for same.

Description of the Related Art

(3) In existing high-dynamic-range imaging (HDR), in most cases, a camera is used to shoot a plurality of images with different exposure values for synthesis. Designed different exposure values are assessed by a basic exposure image as a benchmark, which is a scene detection algorithm. This method is susceptible to some regions that are too dark/too bright in the environment. Input images of different exposures estimated by using the method are brighter/darker, which further affects the tolerance of the synthesis quality, resulting in a relatively high probability of the occurrence of ghosting or shadowing.

BRIEF SUMMARY OF THE INVENTION

(4) According to the first aspect of the disclosure, an image capturing device is provided. The image capturing device includes: a sensor, configured to capture an image of at least one target; a memory, configured to store a plurality of instructions; and a processor, configured to obtain the instructions from the memory to perform the following steps: controlling the sensor to capture a reference image and a processed image; capturing a first bright region and a dark region from the reference image, and capturing a second bright region from the processed image; performing calculations on a first brightness value of the first bright region and a second brightness value of the second bright region respectively with at least two first brightness thresholds, to obtain a first low exposure compensation value and a second low exposure compensation value; and obtaining a high exposure compensation value according to comparisons between a third brightness value of the dark region and at least two second brightness thresholds.

(5) According to the first aspect of the disclosure, an image calculation method is provided. The image calculation method includes: capturing a reference image and a processed image; capturing a first bright region and a dark region from the reference image, and capturing a second bright region from the processed image; performing calculations on a first brightness value of the first bright region and a second brightness value of the second bright region respectively with at least two first brightness thresholds, to obtain a first low exposure compensation value and a second low exposure compensation value; and obtaining a high exposure compensation value according to comparisons between a third brightness value of the dark region and at least two second brightness thresholds.

(6) Therefore, according to the technical content of the disclosure, the image capturing device and the image calculation method described in the embodiments of the disclosure can improve the tolerance of quality of synthetic images, thereby reducing occurrence of ghosting or shadowing.

(7) After referring to the embodiments below, a person of ordinary knowledge in the technical art easily understands the basic spirit and other inventive objectives of the disclosure, as well as the technical means and implementations adopted in the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To make the foregoing and other objectives, features, advantages, and embodiments of the disclosure more comprehensible, the accompanying drawings are described as follows:

(2) FIG. 1 is a schematic diagram of an image capturing device according to an embodiment of the disclosure;

(3) FIG. 2 is a schematic diagram of an image captured by an image capturing device according to an embodiment of the disclosure;

(4) FIG. 3 is a schematic diagram of an image captured by an image capturing device according to

another embodiment of the disclosure;

(5) FIG. 4 is a schematic diagram of an image captured by an image capturing device according to still another embodiment of the disclosure; and

(6) FIG. 5 is a flowchart of an image calculation method according to yet another embodiment of the disclosure.

(7) According to the usual way of working, various features and elements in the drawings are not drawn to scale, and the way of drawing is intended to present the specific features and elements related to the disclosure in the best way. In addition, in different drawings, the same or similar reference signs are used to refer to similar elements/components.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(8) As used herein, “couple” means that two or more elements are in direct physical or electrical contact with each other, or are in indirect physical or electrical contact with each other, or means that two or more elements cooperate or interact with each other.

(9) As shown in FIG. 1, an image capturing device **100** includes a sensor **110**, a memory **120**, and a processor **130**. In terms of connection relationship, the sensor **110** is coupled to the memory **120**, and the memory **120** is coupled to the processor **130**.

(10) To improve the tolerance of quality of synthetic images, and reduce the occurrence of ghosting or shadowing, the disclosure provides the image capturing device **100** shown in FIG. 1. Operations related to the image capturing device **100** are described below in detail.

(11) In an embodiment, the sensor **110** is configured to capture an image of at least one target. The memory **120** is configured to store a plurality of instructions. The processor **130** is configured to obtain the plurality of instructions from the memory **120** to perform the following steps: controlling the sensor **110** to capture a reference image and a processed image; capturing a first bright region and a dark region from the reference image, and capturing a second bright region from the processed image; performing calculations on a first brightness value of the first bright region and a second brightness value of the second bright region respectively with at least two first brightness thresholds, to obtain a first low exposure compensation value and a second low exposure compensation value; and obtaining a high exposure compensation value according to comparisons between a third brightness value of the dark region and at least two second brightness thresholds.

(12) To make the foregoing operations of the image capturing device **100** comprehensible, reference is made to FIG. 2.

(13) As shown in FIG. 1 and FIG. 2, in an embodiment, the processor **130** obtains the plurality of instructions from the memory **120** and controls the sensor **110** to capture a reference image **210** and a processed image **220**. Then, the processor **130** captures a first bright region **310** and a dark region **320** from the reference image **210**, and a second bright region **330** from the processed image **220**. In an embodiment, the processor **130** captures the reference image **210** using the sensor **110**. The reference image **210** may represent an exposure result and imaging information displayed on a current screen, or may be a pre-processed image of HDR. Next, the processor **130** captures the processed image **220** using the sensor **110**. The processed image **220** may be a red, green, and blue (RGB) image generated after reducing image data from an original image file.

(14) Then, the processor **130** performs calculations on a first brightness value of the first bright region **310** and a second brightness value of the second bright region **330** respectively with at least two first brightness thresholds, to obtain a first low exposure compensation value and a second low exposure compensation value. Subsequently, the processor **130** obtains a high exposure compensation value according to comparisons between a third brightness value of the dark region **320** and at least two second brightness thresholds. In an embodiment, the processor **130** performs linear interpolation on the first brightness value of the first bright region **310** and the at least two first brightness thresholds, to obtain the first low exposure compensation value. Then, the processor **130** performs linear interpolation on the second brightness value of the second bright region **330** and the at least two first brightness thresholds, to obtain the second low exposure compensation

value. Further, the processor **130** performs linear interpolation on the third brightness value of the dark region **320** and the at least two second brightness thresholds, to obtain the high exposure compensation value.

(15) In an embodiment, the processor **130** obtains the plurality of instructions from the memory **120** and controls the sensor **110** to perform the following steps: capturing information about a reference brightness of the reference image **210** and information about a processed brightness of the processed image **220**; and creating a first brightness histogram and a second brightness histogram respectively according to the information about the reference brightness and the information about the processed brightness. In an embodiment, the processor **130** collects information about various reference brightnesses in the reference image **210**, and collects statistics from the information about the reference brightnesses, to create the first brightness histogram. Then, the processor **130** collects information about various processed brightnesses in the processed image **220**, and collects statistics from the information about the processed brightnesses, to create the second brightness histogram.

(16) In an embodiment, the processor **130** uses one of the first low exposure compensation value and the second low exposure compensation value as the low exposure compensation value. In an embodiment, if the first low exposure compensation value is 1, and the second low exposure compensation value is 2, the processor **130** selects the value of 1 or 2 as the low exposure compensation value.

(17) In an embodiment, the processor **130** uses a smallest one of the first low exposure compensation value and the second low exposure compensation value as the low exposure compensation value. In an embodiment, if the first low exposure compensation value is 1 and the second low exposure compensation value is 2, the processor **130** selects the value of 1 as the low exposure compensation value.

(18) Referring to FIG. 2 and FIG. 3, in an embodiment, the processor **130** performs a region determination on the reference image **210**, to determine region ratios of the first bright region **310** and the dark region **320** to the reference image **210**. In an embodiment, the processor **130** finds a part with a grayscale brightness of **255** and a part with a grayscale brightness of **0** from the reference image **210** and performs a region determination according to the grayscale brightnesses of **255** and **0**, to determine region ratios of the first bright region **310** and the dark region **320** to the reference image **210**.

(19) In an embodiment, the processor **130** captures a first brightness percentage region from the reference image **210** at a brightest part, where the first brightness percentage region is the first bright region **310**. In an embodiment, a first brightness percentage is 10%. The processor **130** obtains a region with a brightness 10% lower than the highest grayscale brightness of 255 from the reference image **210** as the first bright region **310**.

(20) In an embodiment, the processor **130** captures a second brightness percentage region from the reference image **210** at a darkest part, where the second brightness percentage region is the dark region **320**. In an embodiment, a second brightness percentage is 10%. The processor **130** obtains a region with a brightness 10% higher than the lowest grayscale brightness of 0 from the reference image **210** as the dark region **320**.

(21) In another embodiment, the processor **130** obtains the plurality of instructions from the memory **120** to perform the following steps: performing a region determination on the processed image **220**, to determine a region ratio of the second bright region **330** to the processed image **220**; and capturing a third brightness percentage region from the processed image **220** at a brightest part, where the third brightness percentage region is the second bright region **330**. In an embodiment, the processor **130** finds a part with a grayscale brightness of 255 from the processed image **220** and performs a region determination according to the grayscale brightness of 255, to determine a region ratio of the second bright region to the processed image **220**. Then, the third brightness percentage is 10%. The processor **130** obtains a region with a brightness 10% lower than the highest grayscale

brightness of 255 from the processed image **220** as the third bright region **330**.

(22) Referring to FIG. 2 and FIG. 4, FIG. 4 is a schematic diagram of an image captured by an image capturing device according to still another embodiment of the disclosure. In an embodiment, the processor **130** makes the reference image **210** into a low-exposure image **410** according to the low exposure compensation value. The processor **130** makes the reference image **210** into a high-exposure image **420** according to the high exposure compensation value. In an embodiment, the processor **130** adjusts an exposure value in the reference image **210** according to the low exposure compensation value to make the reference image **210** into the low-exposure image **410**. Then, in an embodiment, the processor **130** adjusts the exposure value in the reference image **210** according to the high exposure compensation value to make the reference image **210** into the high-exposure image **420**.

(23) FIG. 5 is a flowchart of an image calculation method **500** according to yet another embodiment of the disclosure. The image calculation method **500** includes the following steps:

(24) Step **510**: Capture a reference image and a processed image.

(25) Step **520**: Capture a first bright region and a dark region from the reference image, and capture a second bright region from the processed image.

(26) Step **530**: Perform calculations on a first brightness value of the first bright region and a second brightness value of the second bright region respectively with at least two first brightness thresholds, to obtain a first low exposure compensation value and a second low exposure compensation value.

(27) Step **540**: Obtain a high exposure compensation value according to comparisons between a third brightness value of the dark region and at least two second brightness thresholds.

(28) To make the image calculation method **500** comprehensible, reference is made to FIG. 1 to FIG. 5. In step **510**, a processor **130** is configured to capture the reference image **210** and the processed image **220**. Then, referring to step **520**, the processor **130** is configured to capture a first bright region **310** and a dark region **320** from the reference image **210**, and capture a second bright region **330** from the processed image **220**. Then, referring to step **530**, the processor **130** is configured to perform calculations on a first brightness value of the first bright region **310** and a second brightness value of the second bright region **330** respectively with at least two first brightness thresholds, to obtain a first low exposure compensation value and a second low exposure compensation value. Subsequently, referring to step **540**, the processor **130** is configured to obtain a high exposure compensation value according to comparisons between a third brightness value of the dark region **320** and at least two second brightness thresholds.

(29) In an embodiment, referring to step **510**, the processor **130** is further configured to capture information about a reference brightness of the reference image **210** and information about a processed brightness of the processed image **220**, and create a first brightness histogram and a second brightness histogram respectively according to the information about the reference brightness and the information about the processed brightness.

(30) In an embodiment, referring to step **530**, the processor **130** is configured to use one of the first low exposure compensation value and the second low exposure compensation value as the low exposure compensation value.

(31) In an embodiment, referring to step **530**, the processor **130** is configured to use a smallest one of the first low exposure compensation value and the second low exposure compensation value as the low exposure compensation value.

(32) In an embodiment, referring to step **520**, the processor **130** is further configured to perform a region determination on the reference image **210**, to determine region ratios of the first bright region **310** and the dark region **320** to the reference image **210**.

(33) In an embodiment, referring to step **520**, the processor **130** is further configured to capture a first brightness percentage region from the reference image **210** at a brightest part, where the first brightness percentage region is the first bright region **310**, and capture a second brightness

percentage region from the reference image **210** at a darkest part, where the second brightness percentage region is the dark region **320**.

(34) In an embodiment, referring to step **520**, the processor **130** is further configured to perform a region determination on the processed image **220**, to determine a region ratio of the second bright region **330** to the processed image **220**, and capture a third brightness percentage region from the processed image **220** at a brightest part, where the third brightness percentage region is the second bright region **330**.

(35) It should be noted that, a person of ordinary knowledge in the art understands that steps of the image calculation method **500** are drafted according to the implemented functions, are only intended to make the technology in the disclosure more comprehensible, and are not intended to limit the steps. Integrating steps into a same step, splitting a step into a plurality of steps, replacing any step for execution in another step, or adjusting an order of steps still belongs to the implementations in the content of the disclosure. In addition, the image capturing device **100** shown in FIG. **1** to FIG. **4** are only used to describe the steps of the image calculation method **500** exemplarily. However, the image calculation method **500** is not performed by the image capturing device **100** only.

(36) According to the foregoing implementations of the disclosure, application of the disclosure has the following advantages. The embodiments of the disclosure provide an image capturing device and an image calculation method, to improve the tolerance of quality of synthetic images, thereby reducing the occurrence of ghosting or shadowing.

(37) Although the specific embodiments of the disclosure are disclosed in the above implementations, the embodiments are not intended to limit the disclosure. A person of ordinary skill in the art may make variations and modifications without departing from the principle and spirit of the disclosure. Therefore, the protection scope of the disclosure should be subject to the appended claims.

Claims

1. An image capturing device, comprising: a sensor, configured to capture an image of at least one target; a memory, configured to store a plurality of instructions; and a processor, configured to obtain the instructions from the memory to perform the following steps: controlling the sensor to capture a reference image and a processed image; capturing a first bright region and a dark region from the reference image, and capturing a second bright region from the processed image; performing calculations on a first brightness value of the first bright region with at least two first brightness thresholds to obtain a first low exposure compensation value and performing calculations on a second brightness value of the second bright region with another at least two first brightness thresholds, to obtain a second low exposure compensation value; obtaining a high exposure compensation value according to comparisons between a third brightness value of the dark region and at least two second brightness thresholds; and adjusting an exposure value in the reference image according to at least one of: the first low exposure compensation value, the second low exposure compensation value, or the high exposure compensation value.
2. The image capturing device according to claim 1, wherein the processor obtains the instructions from the memory to perform the following steps: capturing information about a reference brightness of the reference image and information about a processed brightness of the processed image; and creating a first brightness histogram and a second brightness histogram respectively according to the information about the reference brightness and the information about the processed brightness.
3. The image capturing device according to claim 1, wherein one of the first low exposure compensation value and the second low exposure compensation value is used as a low exposure compensation value; wherein the processor obtains the instructions from the memory to perform

the following step: adjusting the exposure value in the reference image according to the low exposure compensation value.

4. The image capturing device according to claim 1, wherein a smallest one of the first low exposure compensation value and the second low exposure compensation value is used as a low exposure compensation value; wherein the processor obtains the instructions from the memory to perform the following step: adjusting the exposure value in the reference image according to the low exposure compensation value.

5. The image capturing device according to claim 1, wherein the processor obtains the instructions from the memory to perform the following steps: performing a region determination on the reference image, to determine region ratios of the first bright region and the dark region to the reference image.

6. The image capturing device according to claim 5, wherein the processor obtains the instructions from the memory to perform the following steps: capturing a first brightness percentage region from the reference image at a brightest part, wherein the first brightness percentage region is the first bright region.

7. The image capturing device according to claim 5, wherein the processor obtains the instructions from the memory to perform the following steps: capturing a second brightness percentage region from the reference image at a darkest part, wherein the second brightness percentage region is the dark region.

8. The image capturing device according to claim 1, wherein the processor obtains the instructions from the memory to perform the following steps: performing a region determination on the processed image, to determine a region ratio of the second bright region to the processed image; and capturing a third brightness percentage region from the processed image at a brightest part, wherein the third brightness percentage region is the second bright region.

9. An image calculation method, comprising: capturing a reference image and a processed image; capturing a first bright region and a dark region from the reference image, and capturing a second bright region from the processed image; performing calculations on a first brightness value of the first bright region with at least two first brightness thresholds to obtain a first low exposure compensation value and performing calculations on a second brightness value of the second bright region with another at least two first brightness thresholds, to obtain a second low exposure compensation value; obtaining a high exposure compensation value according to comparisons between a third brightness value of the dark region and at least two second brightness thresholds; and adjusting an exposure value in the reference image according to at least one of: the first low exposure compensation value, the second low exposure compensation value, or the high exposure compensation value.

10. The image calculation method according to claim 9, wherein the step of capturing a reference image and a processed image comprises: capturing information about a reference brightness of the reference image and information about a processed brightness of the processed image; and creating a first brightness histogram and a second brightness histogram respectively according to the information about the reference brightness and the information about the processed brightness.

11. The image calculation method according to claim 9, wherein one of the first low exposure compensation value and the second low exposure compensation value is used as a low exposure compensation value; wherein the image calculation method comprises: adjusting the exposure value in the reference image according to the low exposure compensation value.

12. The image calculation method according to claim 9, wherein a smallest one of the first low exposure compensation value and the second low exposure compensation value is used as a low exposure compensation value; wherein the image calculation method comprises: adjusting the exposure value in the reference image according to the low exposure compensation value.

13. The image calculation method according to claim 9, wherein the step of capturing a first bright region and a dark region from the reference image comprises: performing a region determination

on the reference image, to determine region ratios of the first bright region and the dark region to the reference image.

14. The image calculation method according to claim 13, wherein the step of capturing a first bright region and a dark region from the reference image further comprises: capturing a first brightness percentage region from the reference image at a brightest part, wherein the first brightness percentage region is the first bright region; and capturing a second brightness percentage region from the reference image at a darkest part, wherein the second brightness percentage region is the dark region.

15. The image calculation method according to claim 9, wherein the step of capturing a second bright region from the processed image comprises: performing a region determination on the processed image, to determine a region ratio of the second bright region to the processed image; and capturing a third brightness percentage region from the processed image at a brightest part, wherein the third brightness percentage region is the second bright region.
